

Week 11 – Capstone Progress Documentation

Project ID: PW26_MG_02

Overview

During Week 11, the capstone project progressed through both research and system development activities. The focus of this phase was to strengthen the research contribution of the project while initiating the foundational implementation of the platform architecture. Work carried out during the week included the preparation of intellectual property documentation and the initial setup of backend infrastructure for the stock analytics platform. These developments support the broader objective of building a trustworthy, explainable, and scalable financial intelligence system capable of supporting advanced market analysis and responsible AI-driven financial decision-making.

Progress Update

Development work began on the stock platform infrastructure. The repository was organized with configuration files, environment templates, and container orchestration setup to support modular development. Initial microservices were created for catalog management, pricing, price caching, search indexing, listing APIs, and availability scheduling. Each service was configured with basic server initialization, routing structures, database configuration, and dependency management to establish the platform foundation for future analytics and AI modules.

A major milestone during this week was the submission and filing of a patent titled Explainable Multimodal Financial Reasoning with Affective Signal Fusion for Market Decision Intelligence. This milestone was independently achieved by one student from the project team. The invention proposes an AI-driven framework that integrates quantitative financial indicators, affective signals derived from market communications, and market microstructure analysis within a unified multimodal reasoning architecture. The framework introduces regime-aware signal fusion and explainable decision intelligence to generate transparent predictions and detect potential market manipulation patterns while ensuring interpretability and investor protection.

Upcoming Work

Upcoming work will focus on expanding microservice functionality, implementing data ingestion pipelines, and improving communication between services across the platform. Integration of components from the explainable multimodal reasoning framework will also begin. In addition, security-oriented research will be conducted to strengthen secure data handling, improve API authentication, and identify vulnerabilities to ensure that the platform remains scalable, reliable, and secure.